Installation and Setup Guide

Deep Learning and NLP-Based Real-Time Sign Language Recognition and Contextual Conversion

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1. Introduction

This document describes the step-by-step installation and configuration process for the proposed real-time sign language recognition system. The guide is intended to help users, developers, and evaluators successfully set up the software environment, install required dependencies, and run the application locally. The system is designed as a web-based application using deep learning, computer vision, NLP, and speech synthesis technologies.

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2. System Prerequisites

2.1 Hardware Requirements

- Desktop/Laptop with Intel i3/i5 processor or higher
- Minimum 8 GB RAM (16 GB recommended)
- Webcam (built-in or external RGB camera)
- Minimum 20 GB free disk space
- Optional GPU (NVIDIA) for faster model inference and training

2.2 Software Requirements

- Operating System: Windows 10 / Windows 11
- Python Version: Python 3.10
- Web Browser: Google Chrome / Microsoft Edge (recommended for Web Speech API)
- Code Editor: Visual Studio Code
- Node.js (for frontend execution)

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3. Project Directory Structure

After extracting or cloning the project, the directory structure should be organized as follows:

```
project-root/
│
├── backend/
│   ├── app.py
│   ├── model_server.py
│   ├── requirements.txt
│   └── models/
│       ├── kannada_mobilenetv2_best.pth  [Obtained from .pynb file – colab]
│       └── idx2label.json  [Obtained from .pynb file – colab]
│
├── frontend/
│   ├── src/
│   │   ├── App.jsx
│   │   ├── index.js
│   ├── public/
│   │   ├── index.html
│   └── package.json
```

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4. Backend Installation Steps

4.1 Python Environment Setup

1. Install Python 3.10 from the official Python website.
2. Verify installation using:
3. python --version
4. Navigate to the backend directory:
5. cd backend

4.2 Virtual Environment (Recommended)

Create and activate a virtual environment:

python -m venv venv

venv\Scripts\activate

4.3 Dependency Installation

Install all required Python libraries using:

pip install -r requirements.txt

The key libraries include:

- FastAPI
- OpenCV
- PyTorch
- Torchvision
- NumPy
- Pillow
- gTTS
- Uvicorn

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5. Model Setup

1. Ensure the trained CNN model file (kannada_mobilenetv2_best.pth) is placed inside the models directory.
2. Ensure the label mapping file (idx2label.json) is present.
3. No retraining is required for execution; the model loads automatically during runtime.

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6. Backend Server Execution

Start the FastAPI backend server using:

uvicorn app:app –reload

If successful, the server will run at:

http://127.0.0.1:8000

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7. Frontend Installation Steps

7.1 Node.js Installation

- Install Node.js (LTS version).
- Verify installation:
- node --version
- npm --version

7.2 Frontend Dependency Installation

Navigate to the frontend directory:

cd frontend

npm install

7.3 Running the Frontend

Start the React application:

npm start

The application will open in the browser at:

http://localhost:3000

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8. Web Application Usage

1. Allow webcam access when prompted by the browser.
2. Perform Kannada sign language gestures in front of the webcam.
3. The system displays:
    - Live video feed
    - Predicted Kannada character
    - Confidence score
    - Constructed Kannada text
4. Click the “Speak” option to hear Kannada speech output.

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9. Speech Synthesis Configuration

- Primary: Browser-based Web Speech API (real-time)
- Fallback: Google Text-to-Speech (gTTS)
- Ensure internet connectivity for gTTS audio generation.

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10. Troubleshooting Guidelines

Issue	Possible Cause	Solution
Webcam not detected	Browser permission denied	Enable camera access
No speech output	Browser unsupported	Use Chrome / Edge
Model not loading	Incorrect model path	Verify model directory
Backend not responding	Port blocked	Restart server

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11. Security and Permissions

- Webcam access is used only for real-time gesture recognition.
- No video frames are stored permanently.
- Audio files generated via gTTS are temporary.

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12. Conclusion

This installation guide provides a complete walkthrough for setting up and running the real-time Kannada sign language recognition system. Following these steps ensures smooth deployment, real-time performance, and accurate gesture-to-text and speech conversion. The modular architecture allows future enhancements and easy maintenance.